ChatGPT

Flash Loan Arbitrage & MEV Defense System: Research Landscape, Open Problems, and Publishable Research Agenda

Executive Summary

Flash loans and maximal extractable value (MEV) are closely related but distinct ideas. A **flash loan** is a source of temporary, uncollateralized capital that must be borrowed and repaid within one atomic transaction; if the repayment condition is not met, the entire transaction reverts. Aave V3, for example, exposes both multi-reserve `flashLoan()` and single-reserve `flashLoanSimple()` mechanisms. [1] **MEV**, by contrast, is the extra value obtainable by controlling transaction inclusion, exclusion, or ordering beyond ordinary block rewards and fees. Ethereum deliberately renamed “miner extractable value” to “maximal extractable value” after the transition away from proof-of-work because block-ordering power is no longer fundamentally a mining phenomenon. [2]

The connection is that a flash loan removes an attacker or arbitrageur's capital constraint, while MEV determines whether and where a profitable transaction can be inserted. The historically important bZx attacks demonstrated this interaction: Qin et al. modeled the attacks as optimization problems spanning several DeFi protocols and showed that optimized versions could have produced substantially greater profits than the actual attacks. [3] Beanstalk later demonstrated an even more important lesson: flash loans can amplify **governance** vulnerabilities rather than merely price-oracle or AMM vulnerabilities; the April 17, 2022 attacker used flash-loaned voting power to exploit Beanstalk's governance process and steal roughly $77 million in non-Bean assets. [4]

The research literature has progressed through several phases. *Flash Boys 2.0* established the modern MEV research agenda by documenting arbitrage bots, priority-gas auctions and their consensus implications. [5] Subsequent work quantified sandwiching and arbitrage, synthesized profitable transaction sequences, modeled flash-loan attacks, measured private Flashbots order flow, formally verified DeFi economic security, and studied the post-Merge proposer-builder market. [6] By 2026, the research frontier has shifted toward **private order flow, L2 MEV, encrypted mempools, causal/fair ordering, cross-domain attribution, token-layer MEV and decentralized block building**. [7]

The central conclusion of this survey is that **“hide transactions from the public mempool” is not an adequate definition of MEV defense**. Historical Flashbots measurements found that, in one 2021–2022 observation window, only about 5.6% of identified sandwich attacks used transactions visible in the researcher's public-mempool feed; private transaction channels had already become economically important. [8] Newer evidence continues to show that private routing is not synonymous with MEV immunity: a 2025 study found private sandwich attacks and significant user migration toward private channels after public sandwich exposure, while 2026 work explicitly studies attacks in private L2 mempools. [9]

This leads to the strongest publishable opportunity: rather than building only a “flash-loan arbitrage bot,” build a **counterfactual MEV defense research platform** that asks, for every proposed transaction:

> *What is the maximum economically plausible harm to this transaction under alternative ordering, information-leakage and capital assumptions—and which defense minimizes that harm at acceptable latency and cost?*

Such a system would combine an EVM state simulator, flash-loan-aware attack synthesis, historical and live order-flow replay, adversarial block construction, counterfactual user-welfare measurement, and adaptive routing or ordering defenses. It can be developed safely using historical state and isolated devnets rather than exploiting live protocols.

Among the research gaps examined below, the highest-impact/highest-feasibility problems are: **counterfactual per-transaction MEV-risk estimation; compositional flash-loan vulnerability discovery; systematic testing of private/encrypted-order-flow leakage; and economically rigorous comparison of MEV-defense mechanisms**. Cross-chain/CEX–DEX attribution is potentially even more important scientifically, but is considerably harder because the necessary off-chain timing and order-book data are fragmented.

The best primary project recommendation is therefore **Adaptive Counterfactual MEV Defense**, with **Flash-Loan-Aware Economic Attack Synthesis** as a complementary security-analysis component. A realistic ten-month project can produce a reproducible historical dataset, a local builder/relay test environment, several baseline reproductions, a prototype detector/defense, adversarial experiments and a conference-quality paper. Flashbots' `rbuilder`, Decker/Builder Playground, Artemis, public Protect/MEV-Share datasets and normal Ethereum execution traces provide enough open infrastructure to avoid constructing the whole stack from scratch. [10]

Foundations and System Model

What a flash loan really changes

Suppose a normal arbitrage requires capital q. Without a flash loan, the arbitrageur must already own or borrow q through a conventional collateralized mechanism. With a flash loan, the capital leg becomes approximately:

$$\text{borrow } q \rightarrow \text{perform actions} \rightarrow \text{receive } q + \Pi \rightarrow \text{repay } q + \text{premium}$$

inside one atomic blockchain transaction. Aave's implementation requires the borrowed amount plus the applicable premium to be returned—or, for supported `flashLoan()` modes, converted into an authorized debt position—before successful completion; otherwise execution fails/reverts. [11]

An arbitrage contract is therefore economically interested in

$$\Pi = V_{\text{final}} - V_{\text{initial}} - F_{\text{DEX}} - F_{\text{flash}} - G_{\text{gas}} - B_{\text{builder}}$$

where B_{builder} represents an explicit or implicit payment for favorable block inclusion/order.

For a constant-product AMM,

$$xy = k,$$

a sufficiently large trade changes the pool's reserves and therefore its marginal exchange price. If two pools imply different prices for the same asset, an arbitrageur can potentially trade against the cheap pool and reverse the position against the expensive pool until fees, price impact and execution costs remove the opportunity. The important research issue is not simply whether a positive mathematical spread exists but whether the **entire atomic state transition remains profitable under adversarial competition and realistic block construction**. Work on automatic arbitrage discovery, formal AMM MEV maximization, and large-scale Ethereum arbitrage measurement all demonstrates the importance of modeling execution rather than merely comparing quoted prices. [12]

Arbitrage, sandwiching, and economic attacks are not equivalent

A useful taxonomy is:

Activity	Mechanism	Typical externality
Backrun arbitrage	Trade after a price-moving event to restore cross-market consistency	Can improve price alignment, although competition transfers surplus to builders/validators
Cyclic arbitrage	Execute a profitable token cycle across one or more DEXs	Usually searcher competition and gas/blockspace consumption
Liquidation	Trigger an undercollateralized position and collect its incentive	Necessary to many lending protocols, but highly competitive
Sandwich	Buy before a victim, let the victim move the price, sell immediately after	Victim receives systematically worse execution
Oracle manipulation	Temporarily move a price source and use that distorted state elsewhere	Can cause lending/minting/liquidation losses
Governance manipulation	Temporarily acquire voting power and execute a malicious governance transition	Protocol control or treasury loss
Generalized front-running	Observe another profitable transaction and construct a competing transaction	Copies/appropriates another actor's opportunity

Flash Boys 2.0 documented the fee-bidding competition around DEX arbitrage and called attention to its consensus-security implications. [5] Zhou et al. subsequently formalized sandwich execution and empirically measured high-frequency adversarial trading on Ethereum DEXs. [13] Qin, Zhou and Gervais measured sandwiching, liquidations and arbitrage over 32 months and estimated $540.54 million of blockchain-extractable profit across the activities and datasets they considered. [14]

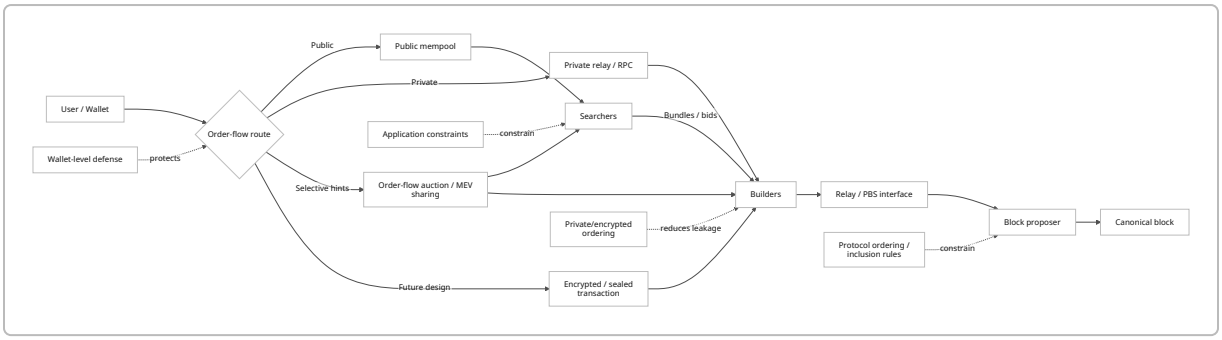
A flash loan should consequently be treated as a **capital primitive**, not itself as a vulnerability. The economically dangerous combination is usually:

$$\text{cheap atomic capital} + \text{manipulable state} + \text{composable protocols} + \text{profitable terminal state}.$$

That distinction is visible in both bZx, where flash liquidity interacted with trading/oracle dependencies, and Beanstalk, where it interacted with governance. [15]

Where MEV comes from in today's transaction pipeline

A simplified Ethereum-style flow is:



Ethereum's current MEV architecture includes out-of-protocol proposer-builder separation, where validators can use MEV-Boost to solicit blocks from specialized builders via relays. Flashbots describes MEV-Boost as middleware that allows validators to access a competitive block-building market, while Ethereum's roadmap is pursuing in-protocol proposer-builder separation to remove this trusted middleware architecture. [16] EIP-7732 specifies a concrete enshrined-PBS design, but as of September 2026 its EIP page still describes it as undergoing peer review rather than something that should be assumed to be part of today's universally deployed execution model. [17]

The architectural trend is therefore away from the old mental model

$$\text{user} \to \text{public mempool} \to \text{miner}$$

toward a multi-actor market involving wallets, RPCs, searchers, private order-flow providers, builders, relays and validators. Flashbots' current BuilderNet initiative adds another model: decentralized collaborative block building inside trusted execution environments, with an open-source refund rule intended to redistribute value contributed by order-flow providers. [18]

What a research-grade defense system should optimize

A useful defense objective is not “eliminate all MEV.” Some MEV, such as arbitrage and liquidation, provides economically useful functions. Instead define counterfactual user harm:

$$H(tx) = U(tx, \sigma_{\text{benign}}) - \min_{\sigma \in \Sigma_{\text{plausible}}} U(tx, \sigma),$$

where U is the user's execution utility and $\Sigma_{\text{plausible}}$ is the set of orderings an economically realistic adversary or builder can produce.

A defense d can then be evaluated with

$$J(d) = H_d + \lambda_1 C_d + \lambda_2 L_d + \lambda_3 R_d + \lambda_4 Z_d,$$

where C is monetary execution cost, L latency, R revert/failure probability and Z a decentralization or trust penalty.

This formulation is particularly important because research on Uniswap execution shows that observed "slippage" contains both benign market movement and adversarial components; simply measuring quoted-price versus realized-price difference does not identify MEV harm. [19]

Prior Work and Ecosystem Survey

Academic literature

The literature below is prioritized around original measurement, formalization, attack synthesis and defense work rather than surveys.

Work	Core method and dataset	Main contribution / finding
Flash Boys 2.0, Daian et al., IEEE S&P 2020	Empirical monitoring of DEX arbitrage bots and priority-gas auctions; game-theoretic model of fee-bidding competition	Established the modern MEV research agenda; showed bots competing through latency and fees and argued that sufficiently large extractable rewards can create consensus-level instability. [5]
Attacking the DeFi Ecosystem with Flash Loans for Fun and Profit, Qin et al., FC 2021	Reconstructs the two February 2020 bZx attacks; optimization over state/actions spanning six DeFi protocols	Systematic analysis of flash-loan attacks; reported actual attack profits around $350K and $600K and optimized strategies of roughly $829.5K and $1.1M, demonstrating how atomic composability amplifies economic attack surfaces. [3]
High-Frequency Trading on Decentralized On-Chain Exchanges, Zhou et al., IEEE S&P 2021	Analytical sandwich model, Ethereum/Uniswap measurements and competing-adversary simulations	Derives sandwich profitability and studies dependence on victim position and searcher competition; empirically demonstrated that a single adversary could earn thousands of dollars per day in the measured historical environment. [13]
On the Just-In-Time Discovery of Profit-Generating Transactions in DeFi Protocols, Zhou et al., IEEE S&P 2021	`DEFIPOSER-ARB` graph search/ Bellman-Ford and `DEFIPOSER-SMT` logical solving; 150 days, 96 protocol actions and 25 assets	Automatically synthesizes profitable DeFi transactions. The SMT version rediscovered the bZx-style opportunity, illustrating how an economic exploit can emerge from composition rather than an ordinary code bug. [20]

Work	Core method and dataset	Main contribution / finding
Quantifying Blockchain Extractable Value: How Dark Is the Forest?, Qin, Zhou & Gervais, IEEE S&P 2022	32-month measurement of sandwiches, arbitrage and liquidations; 11,289 addresses, 49,691 cryptocurrencies and 60,830 markets	Quantified $540.54M of measured profit and analyzed generalized transaction copying/front-running and the consensus incentives created by very large opportunities. [14]
Maximizing Extractable Value from Automated Market Makers, Bartoletti et al., FC 2022	Formal AMM semantics and construction of value-maximizing adversarial strategies	Provides an important theoretical baseline: instead of heuristically looking for attacks, derive strategies that maximize value extractable from a given set of AMM user transactions. [21]
A2MM, Zhou, Qin & Gervais	Automated arbitrage market maker that internalizes cross-AMM arbitrage	Proposes turning arbitrage into protocol/user revenue rather than an external searcher race; its evaluation estimated a 32.8% reduction in AMM-related blockspace usage and substantial potential fee reductions under the studied assumptions. [22]
A Flash(bot) in the Pan: Measuring MEV in Private Pools, Weintraub et al.	Flashbots API history through block 14,444,725: about 1.196M Flashbots blocks and 3.249M bundles; independent pending-transaction capture to distinguish public/private inclusion	Demonstrated the measurement blind spot created by private pools. In a four-month subset, 81.15% of identified sandwiches used Flashbots and only 5.6% of overall sandwiches were visible through the authors' public pending-transaction feed. [8]
Clockwork Finance, Babel et al., IEEE S&P 2023	Formal verification framework modeling Uniswap, Uniswap V2, SushiSwap, MakerDAO plus primitives such as flash loans, voting and airdrops	Introduces a compositional notion of extractable value and attack-exhaustive economic analysis; experiments found large quantities of previously latent extractable value without hard-coding individual attack strategies. [23]
A Large Scale Study of the Ethereum Arbitrage Ecosystem, McLaughlin et al., USENIX Security 2023	28 months; detector covers more DEX applications and sophisticated price functions than predecessors	Identified 3.8M arbitrages yielding $321M and approximately 4B potential opportunities; also found that some opportunities persisted for multiple blocks and that a growing share of arbitrage revenue was paid to block producers. [24]

Work	Core method and dataset	Main contribution / finding
Time to Bribe: Measuring Block Construction Markets, Wahrstätter et al.	Longitudinal post-Merge PBS measurement from September 2022 to May 2023; builder/relay concentration and proposer payments	Shows that MEV evolved into a block-construction market rather than disappearing after proof-of-stake; studies concentration, payments and relationships between periods of market stress and MEV bids. [25]
Don't Let MEV Slip	Compares Uniswap-interface quotes and realized execution for USDC–ETH and PEPE–ETH pools and decomposes slippage	Important methodological shift from "searcher revenue" to **user execution quality**, separating benign slippage from adversarial extraction. [19]

Three 2026 developments deserve special attention because they directly define the frontier rather than simply repeating older Ethereum-mainnet analysis.

First, *More to Extract: Discovering MEV by Token Contract Analysis* argues that existing MEV-discovery tools typically focus on application-layer pools or attacker contracts and can miss MEV originating in non-standard token semantics. Its pipeline combines static and dynamic analysis of token-supply-control functions; the authors explicitly demonstrate cases outside Nyx's application-layer model and conduct experiments on historical state rather than live exploitation. [26]

Second, *Time Is Money: Incentivized Causal Transaction Ordering* proposes PRECEDE, a mechanism where transactions are prioritized by a super-linearly bid-weighted randomized lottery rather than simple highest-bid-first ordering. The authors' game-theoretic argument is that a user's deterring bid can form an equilibrium in which rational attackers do not front-run; they also reduce sandwich resistance to the front-running problem. This is a July 2026 preprint and should therefore be treated as emerging rather than established consensus. [27]

Third, the 2026 SoK on the evolution of MEV emphasizes that the problem is expanding beyond a single-chain Ethereum mempool into cross-chain and cross-domain settings and identifies standardized metrics and detection benchmarks as continuing research needs. [28] Contemporary work on the **origins** of MEV likewise shifts the question from "who extracted value?" to "what state perturbation created the opportunity?", an extremely useful distinction for causal attribution and defense. [29]

Open-source projects and research tooling

Project	Purpose	Main language	License	Maturity as of Sept. 2026	Repository
MEV-Boost	Validator middleware implementing today's out-of-protocol builder/relay market	Go	MIT	Production infrastructure widely intended for validator operation. [30]	flashbots/mev-boost
rbuilder	High-performance Ethereum block builder; supports historical backtesting and live relay submission	Rust	MIT / Apache-2.0	Flashbots states that it has run in production since Q1 2024 and remains under active development. [31]	flashbots/rbuilder
Artemis	Modular event → strategy → action/executor framework for MEV bots	Rust	MIT / Apache-2.0	Strong research/reference framework; includes MEV strategy examples rather than being a turnkey profitable bot. [32]	paradigmxyz/artemis
MEV-Share Node	Matchmaking infrastructure for selectively disclosed order flow and backruns	Go	AGPL-3.0-or-later	Useful protocol/reference implementation for MEV-sharing experiments. [33]	flashbots/mev-share-node
simple-blind-arbitrage	Demonstrates blind backrunning against MEV-Share hints; includes an on-chain two-pool Uniswap-V2-style optimizer	Solidity + JavaScript/Foundry tooling	MIT	Educational/reference implementation, not a production bot. [34]	flashbots/simple-blind-arbitrage

Project	Purpose	Main language	License	Maturity as of Sept. 2026	Repository
mev-inspect-py	Historical extraction and classification of swaps, arbitrage, miner payments and profits	Python	MIT	Explicitly deprecated; still valuable for reproducing older MEV measurement methodology. [35]	flashbots/mev-inspect-py
mev-geth	Historical Flashbots/MEV-Auction modifications to the Go Ethereum client	Go	LGPLv3 library / GPLv3 command components	Legacy proof-of-work-era infrastructure; valuable historically, not the right foundation for a new post-Merge project. [36]	flashbots/mev-geth
Builder Playground	Reproducible local Ethereum L1/L2 builder, relay, EL and CL development environments	Go	MIT	Stable but no longer actively maintained; Flashbots now recommends its successor Decker for new work. [37]	flashbots/builder-playground
Decker	Composable local devnets suitable for builder/relay experiments and CI	TypeScript/ Deno	MIT	Current successor to Builder Playground and a strong environment for controlled MEV experiments. [38]	flashbots/decker

For a new research project, the best combination is **rbuilder + Decker + an EVM fork/simulator + Python/ Rust analysis code**. `rbuilder` already provides builder-level control and backtesting; Decker makes isolated network experiments reproducible; Artemis provides a clean architecture for agent/searcher implementations. [10]

An important reproducibility warning is that many online “MEV bot” repositories are educational, obsolete, incomplete, or outright unsafe. A scientifically defensible project should therefore base its infrastructure on maintained protocol/client repositories and derive experimental bots from a clearly specified model rather than treating anonymous “arbitrage-bot” GitHub code as ground truth.

Industry mechanisms and protocol proposals

Flashbots MEV-Boost and MEV-Share. MEV-Boost externalized Ethereum block construction into a builder market. MEV-Share subsequently introduced selective transaction disclosure: searchers can receive hints instead of the entire user transaction and compete to construct backruns, with part of the resulting value intended to return to the originating order flow. [39] Flashbots has also published historical Protect and MEV-Share data, making these systems unusually attractive for empirical research. [40]

BuilderNet. Flashbots' current direction is a TEE-backed collaborative block-building network designed to reduce dependence on exclusive order-flow agreements and redistribute contributed MEV using an open-source refund rule. Its first release dates to November 2024, according to Flashbots' current project description. [18] This is important because the defense problem shifts from “can I trust a relay not to leak my transaction?” to a wider system question involving TEE integrity, operator diversity, attestation, privacy and refund-mechanism incentives.

Enshrined PBS. EIP-7732 moves proposer-builder exchange into Ethereum consensus and separates consensus-block and execution-payload timing. Its stated motivation includes eliminating trusted middleware for block-construction delegation and improving the block-validation critical path. [17] Enshrined PBS mitigates some infrastructure risks but does **not**, by itself, make harmful MEV disappear: specialized builders can still optimize ordering.

FOCIL. EIP-7805 proposes Fork-choice Enforced Inclusion Lists, where a committee publishes lists of transactions and attesters refuse to support a block that improperly omits valid listed transactions. The proposal explicitly responds to censorship concerns created by concentrated specialized block builders. [41] FOCIL therefore primarily addresses **inclusion/censorship**, not confidentiality or sandwich resistance.

LUCID encrypted mempool. EIP-8184 proposes carrying sealed transactions through Ethereum's public inclusion pipeline while withholding their contents until the scheduling decision is committed. Particularly revealing is the EIP's own motivation: it argues that private order-flow systems protect transactions at the cost of opacity and trusted relationships, while Ethereum does not yet have a single threshold-encryption construction satisfying all desired scale, setup, ciphertext-size, CCA-security and post-quantum properties. [42] This is effectively an official statement of a major open research problem.

Major incidents and what they teach

Incident	Mechanism	Research lesson
bZx, February 2020	Flash loans composed lending, margin trading, DEX liquidity and manipulated state in one transaction	Economic vulnerabilities may exist only across protocol boundaries. Qin et al.'s optimization showed considerably more profit was theoretically available than the attackers actually captured. [3]

Incident	Mechanism	Research lesson
Harvest Finance, October 2020	Flash-liquidity-assisted manipulation of the USDC/USDT pricing environment around Curve Y pool interactions and Harvest vault accounting	A protocol can be individually “correct” at the code level yet insecure if it trusts a manipulable instantaneous economic state. [43]
Beanstalk, April 17, 2022	Flash-loaned assets gave the attacker the governance power needed to execute a malicious governance path	Flash loans are not merely a DEX/oracle problem; any “balance now = legitimate long-term economic stake” assumption is suspect. Approximately $77M of non-Bean assets was stolen. [4]
Flashbots relay exploit, April 3, 2023	A malicious proposer induced an ultra sound relay to reveal a block body and then reused transactions to break searchers' sandwich bundles	MEV infrastructure itself becomes a security-critical protocol. Flashbots reported roughly $20M stolen from searchers and patched the relay behavior. [44]
Jito mempool suspension, March 8, 2024	A low-latency/private ordering infrastructure became an avenue for negative user externalities and sandwiching	MEV defense and MEV infrastructure can create the very adversarial market they seek to manage. Jito officially suspended its mempool because of negative externalities affecting Solana users and has since added explicit sandwich-mitigation mechanisms. [45]

The consistent pattern is that **capital availability, information availability and ordering power interact**. Defending only one of the three leaves exploitable combinations of the other two.

Research Gaps and Open Problems

Prioritized gap matrix

The following prioritization is my assessment of research value rather than a claim from any one source.

Priority	Open problem	Dimension	Impact	Feasibility in 6–12 months	Why unresolved
Very high	Counterfactual per-transaction MEV harm estimation	Technical / economic	Very high	High	Current systems typically label observed attacks; they rarely estimate the worst plausible outcome that *could* have happened under alternative ordering
Very high	Flash-loan-aware compositional vulnerability synthesis	Security / technical	Very high	High–medium	Search spaces explode across protocol, token, oracle, governance and temporal state
Very high	Private/ encrypted-order-flow leakage measurement	Security / privacy	Very high	Medium–high	“Private” channels still expose metadata, hints, timing or privileged endpoints
High	Adaptive MEV defense selection	Systems / economic	High	High	No single defense dominates for swaps, arbitrage, liquidation, minting, governance and latency-sensitive transactions
High	MEV-defense mechanism economics	Game theory	High	Medium	Protecting users can shift rents or concentration rather than eliminating extraction
High	Cross-domain MEV provenance	Measurement / economics	Very high	Medium–low	Relevant cause and profit may span CEXs, L1s, L2s, bridges and multiple blocks
Medium–high	Defense auditability and policy taxonomy	Policy / systems	Medium–high	High	Regulators need operational definitions separating harmful extraction, market-making, arbitrage and protocol-essential activity

Technical and security gaps

Observed MEV is not the same as available MEV. Existing measurement systems generally infer extraction after the fact from transactions that landed on-chain. DEFIPOSER, formal AMM optimization and Clockwork go further by searching possible strategies, but general counterfactual optimization over realistic private order flow, builder policies and transaction dependencies remains computationally difficult. [46] A good defense should estimate both

$$MEV_{\text{observed}} \quad \text{and} \quad MEV_{\text{available but unrealized}}.$$

Composability has outgrown protocol-level abstractions. Clockwork demonstrated the value of compositional economic models, but 2026 token-analysis work explicitly shows that tools concentrating on application contracts can overlook profit opportunities generated by non-standard token behavior or runtime-only dependencies. [47] A genuine flash-loan defense therefore cannot reduce the system to "AMM pools + oracle." It should reason across tokens, proxies, governance modules, hooks, callbacks, lending markets and dynamically created state dependencies.

Private does not mean secret enough. Historical Flashbots measurements already exposed the methodological problem of private transactions being invisible to conventional mempool datasets. [8] More recent empirical work reports sandwiching through private Ethereum channels, while 2026 work extends the question to private L2 mempools. [9] A defense therefore needs an explicit leakage model:

$$L = \{\text{sender}, \text{nonce}, \text{target}, \text{amount range}, \text{timing}, \text{gas}, \text{hint}, \text{route}, \ldots\}.$$

A system should be evaluated against an attacker who receives *some subset of* L, not only the unrealistic extremes "whole transaction visible" versus "nothing visible."

Encrypted mempools have unresolved availability and cryptographic trade-offs. LUCID explicitly recognizes that Ethereum does not yet have a single cryptographic mechanism satisfying every desired property at protocol scale. [42] Encryption creates additional problems around key release, non-reveal, denial of service, metadata leakage, censorship and the ability to validate transactions before content is available. This is fertile systems/security research.

Economic and game-theoretic gaps

Minimizing searcher profit is not necessarily equivalent to maximizing user welfare. Arbitrage often improves market price consistency, and liquidation keeps lending systems solvent; aggressively suppressing all MEV could make markets less efficient. Conversely, *Flash Boys 2.0*, the large-scale arbitrage study and post-Merge block-market measurements show that competition can create bidding races and transfer a large fraction of available surplus toward block producers rather than users. [48]

The right optimization problem is closer to

$$\max(W_{\text{users}} + W_{\text{LPs}} + W_{\text{protocol}} + W_{\text{validators}}) - C_{\text{latency}} - C_{\text{concentration}} - C_{\text{attack}},$$

subject to security and censorship-resistance constraints.

Defense adoption itself is a game. Suppose protected order flow gives users better prices but deprives public searchers of information. Searchers may migrate toward privileged builders; builders with superior private flow may win more blocks; wallets then route toward those builders because they have better execution. This can produce a reinforcing loop:

$$\text{more private order flow} \rightarrow \text{better builder bids} \rightarrow \text{more blocks} \rightarrow \text{more wallet flow}.$$

FOCIL's motivation explicitly identifies concentration among sophisticated builders as a censorship-resistance concern. [41] BuilderNet is an industry attempt to change the economics of exclusive order flow rather than merely conceal individual transactions. [18] Measuring whether such architectures truly reduce concentration is still an open empirical problem.

Fair ordering is not obviously incentive compatible. Arrival-time ordering is difficult in a geographically distributed system because no globally objective reception time exists. Random ordering can create spam incentives. Encryption can shift attacks to metadata or post-decryption races. PRECEDE's authors argue that earlier ordering defenses do not fully enforce causality and therefore introduce an explicitly game-theoretic randomized bidding mechanism. [27] Whether such mechanisms remain robust with heterogeneous latency, colluding builders, private flow and multi-transaction bundles is a promising unanswered question.

Cross-domain and attribution gaps

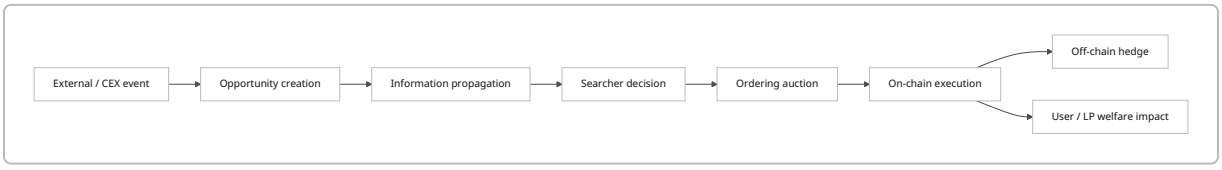
Atomic Ethereum arbitrage is now only one part of the phenomenon. Cross-chain SoK work and newer attribution research emphasize that an opportunity's **creator**, **observer**, **extractor**, **settlement venue** and **ultimate hedge venue** need not be the same actor or chain. [49]

For example:

$$\text{CEX price shock} \rightarrow \text{L2 DEX stale price} \rightarrow \text{L2 arbitrage} \rightarrow \text{L1 settlement}$$

cannot be properly characterized from one Ethereum transaction trace.

A publishable open problem is therefore to create a causal provenance graph:



This separates **the origin of MEV** from **the actor who captured it**.

Policy and governance gaps

The UK Financial Conduct Authority's MEV research note describes MEV and blockchain oracles as emerging technical features that introduce potential risks and vulnerabilities and states that the work is intended to build regulatory understanding. [50] The notable point is what is *not* yet standardized: there is no universal operational rule by which every arbitrage, backrun, liquidation, sandwich, builder rebate or private-order-flow agreement is mechanically classified as beneficial, harmful or market-abusive.

A useful policy research contribution would therefore avoid arguing “MEV should be banned” or “MEV is always legitimate.” Instead it could produce measurable attributes:

$$(\text{information asymmetry}, \text{victim harm}, \text{price improvement}, \text{ordering privilege}, \text{disclosure}, \text{conflict of interest}, \text{concen}$$

and show how real transactions distribute across this feature space. That would give policy discussions reproducible technical evidence rather than taxonomy by anecdote. The FCA's continued use of MEV research to inform regulatory expertise makes this an unusually practical academic-policy intersection. [50]

Concrete Research Directions

Adaptive Counterfactual MEV Defense

Research question. Can the system predict, *before submission*, the economically realistic worst-case execution of a transaction and automatically choose a defense that produces better expected user welfare than any fixed routing policy?

Hypothesis.

> A transaction-specific counterfactual risk score combined with adaptive routing will reduce adversarial execution loss significantly relative to always-public, always-private and static-slippage baselines, without materially increasing failed transactions or tail latency.

Novelty

Most MEV monitoring starts from an attack that actually happened. This direction starts with an unconfirmed transaction tx and constructs adversarial counterfactual blocks:

$$B_1, B_2, \ldots, B_n$$

under public visibility, partial hints, private routing, competing searchers and different builder objectives.

Define:

$$Risk(tx) = \max_i \left[U(tx, B_{\text{benign}}) - U(tx, B_i) \right].$$

The system then selects among defenses such as public submission, protected RPC, selective-hint auction, tighter application constraints, batch execution, delayed execution or encrypted/sealed routing.

The novelty is not another sandwich classifier. It is a **pre-execution adversarial risk oracle plus policy engine**.

Methodology

Build four components:

State reconstruction. Replay historical Ethereum blocks and reconstruct pool/token/protocol state immediately before selected user transactions.

Counterfactual builder. Use `rbuilder` or a custom EVM block builder to insert/reorder competing transactions, generate legal front/backrun structures, and model multiple searchers. `rbuilder` directly supports historical block-building experiments and is therefore a particularly good foundation. [31]

Risk estimator. Estimate transaction sensitivity using features such as trade size relative to liquidity, slippage tolerance, pool topology, price impact, potential cyclic routes, observable metadata and external volatility.

Defense selector. Optimize expected utility across candidate channels.

Experimental baselines should include a simple sandwich detector, fixed private routing, fixed public routing and a price-impact-only risk rule. The evaluation can use historical periods with known sandwiches and arbitrage plus synthetic counterfactual adversaries.

Evaluation metrics

Primary:

$$\text{Prevented Harm Ratio} = \frac{H_{\text{baseline}} - H_{\text{defense}}}{H_{\text{baseline}}}.$$

Also report attack recall, false-positive rate, realized execution price, price improvement, gas cost, builder/searcher payment, revert rate, added latency, p95/p99 latency and calibration error between predicted and realized counterfactual harm.

A particularly strong paper would report **regret**:

$$Regret = U(d^*) - U(\hat{d}),$$

where d^* is the best defense in hindsight and $\hat{d}$ is the defense chosen online.

Required skills

Rust or Go for high-performance builder integration; Python for empirical analysis and causal/statistical modeling; Solidity/EVM internals; AMM mathematics; Ethereum transaction tracing; basic mechanism design.

Expected challenges

The hardest problem is ground truth: one observes only the canonical block, not every possible ordering. Counterfactual simulation must therefore make attacker assumptions explicit rather than pretending to measure a unique “true MEV.”

Another difficulty is private order-flow data. Historical Protect/MEV-Share datasets published by Flashbots can partially close this gap, but they cannot represent the entire private market. [40]

Publication targets

Best fits: **USENIX Security, IEEE S&P, ACM CCS** if the emphasis is adversarial systems/security; **Financial Cryptography** or **ACM AFT** if the emphasis is DeFi economics and mechanism design; **ACM IMC** if the main contribution is measurement infrastructure.

Overall assessment: highest feasibility and the best main project.

Flash-Loan-Aware Economic Attack Synthesis

Research question. Can automated analysis find attacks in which the code is locally correct but an attacker uses flash liquidity and protocol composition to reach an economically unsafe state?

Hypothesis.

> A hybrid graph + symbolic/constraint + dynamic-state search that explicitly treats flash liquidity as an unconstrained temporary capital primitive will discover multi-protocol economic attack paths more efficiently and more completely than application-only analyzers.

Novelty

DEFIPOSER showed graph and SMT approaches can discover profitable transaction sequences, while Clockwork generalized compositional economic security. [51] The 2026 token-analysis work demonstrates a complementary blind spot: MEV can originate from token-level semantics that application-focused tools ignore. [26]

A new tool could combine these ideas with a richer state graph:

$$G = (V, E)$$

where vertices include:

- assets and balances,
- lending positions,
- AMM reserves,
- oracle observations,
- token-supply transformations,
- governance voting weight,
- protocol permissions,

and edges are callable state transitions.

A flash-loan edge introduces temporary liquidity:

$$balance_a \leftarrow balance_a + q$$

subject only to a terminal repayment constraint:

$$balance_a^{end} \geq q + fee.$$

This transforms attack discovery into constrained search for any terminal state where

$$V_{\text{attacker}}^{end} - V_{\text{attacker}}^{start} - costs > 0.$$

Methodology

Create a benchmark suite from historical incidents including bZx, Harvest and Beanstalk, plus non-vulnerable protocol snapshots as negative controls. [52]

Implement progressively stronger analyzers:

1. graph-only arbitrage search,
2. protocol-level SMT search,
3. flash-liquidity-aware SMT,
4. token/oracle/governance-aware dynamic analysis,
5. combined beam search or learned search heuristic.

Execute all synthesized candidates only against historical forks or isolated devnets.

Metrics

Attack coverage; time-to-first-exploit; total state transitions explored; false positives on non-vulnerable versions; maximum synthesized profit; minimum external capital required; number of protocols crossed; solver time and memory; exploit reproducibility across independent forks.

A compelling metric is:

$$Coverage(k) = \frac{\text{historical exploit classes rediscovered within } k \text{ minutes}}{\text{total benchmark classes}}.$$

Required skills

Solidity/EVM, program analysis, symbolic execution or SMT, graph algorithms, DeFi protocol mechanics and security testing.

Challenges

State-space explosion is severe. The most interesting economic vulnerabilities involve long sequences of individually legitimate operations. Dynamic callbacks, proxies, custom tokens and oracle timing can invalidate simple symbolic abstractions.

Publication targets

IEEE S&P, USENIX Security, ACM CCS, NDSS, or **Financial Cryptography**. A strong open benchmark alone could also become a highly reusable research artifact.

Overall assessment: strongest pure-security direction and excellent companion to the adaptive-defense system.

Leakage-Resilient Private and Encrypted Order Flow

Research question. How much information can an attacker learn before execution under apparently "protected" transaction-routing architectures, and how much MEV remains?

Hypothesis.

> Most practical private-order-flow mechanisms have a measurable residual leakage surface; a metadata-aware sealed-order architecture can substantially reduce exploitable information while preserving acceptable inclusion latency and failure rates.

Why this is genuinely open

Historical measurements already show that the public mempool represents only a subset of economically important ordering activity. [8] Private sandwiching has subsequently been observed empirically, and private L2 settings are now an explicit research target. [9] At the same time, LUCID acknowledges that a fully satisfactory Ethereum-scale encrypted mempool construction remains cryptographically difficult. [42]

Methodology

Construct a controlled Decker/`rbuilder` devnet and define leakage profiles:

$$L_0 = \varnothing$$

$$L_1 = \{\text{arrival time}\}$$

$$L_2 = \{\text{arrival}, \text{sender}\}$$

$$L_3 = \{\text{arrival}, \text{target contract}, \text{gas}\}$$

$$L_4 = \{\text{MEV-Share-style hints}\}$$

$$L_5 = \{\text{full plaintext transaction}\}.$$

Train or program adversarial searchers to infer whether a profitable sandwich/backrun exists from each profile.

Compare public mempool, private relay, selective disclosure and LUCID-like commit-before-reveal prototypes.

Attack not only the transaction plaintext but:

- sender/linkability,
- nonce sequences,
- transaction size,
- arrival-time clustering,
- endpoint behavior,
- builder acknowledgment timing,
- repeated wallet routing,
- failed-decryption/reveal behavior.

Metrics

Mutual information between leakage and hidden transaction class; attack success rate; extracted value; victim execution loss; anonymity-set size; classification AUC; inclusion probability; decryption failure rate; additional bytes; CPU/network cost; p95 end-to-end latency.

A particularly publishable metric would be the **MEV leakage frontier**:

$$(\text{privacy leakage}, \text{latency}, \text{user loss})$$

for each system.

Required skills

Distributed systems, Ethereum builders, cryptographic protocols, network measurement, information theory, Rust/Go and statistical analysis.

Challenges

Real private relay internals are not fully observable; local simulation must carefully distinguish properties intrinsic to an architecture from artifacts of the test harness.

Cryptography can also overwhelm the project. A good first paper can evaluate **interfaces and leakage properties** without inventing a new threshold-encryption primitive.

Publication targets

USENIX Security, NDSS, ACM CCS, IEEE S&P, and potentially **PETS** if the contribution is strongly privacy-oriented.

Overall assessment: high novelty and excellent current relevance.

Mechanism Design for User-Rebating MEV Markets

Research question. Can a block-building/order-flow mechanism return more surplus to users without reducing price efficiency or creating excessive builder concentration?

Hypothesis.

> An appropriately designed combination of user rebates, non-exclusive order flow and randomized or constrained ordering can improve aggregate user welfare while maintaining validator revenue and reducing concentration relative to simple first-price MEV auctions.

Motivation

The early MEV literature identified gas/fee bidding wars. [5] Post-Merge work showed the same economics migrating into builder bids and block-construction markets. [25] BuilderNet explicitly attempts to alter this equilibrium by redistributing value contributed by order flow, while PRECEDE attacks front-running incentives through randomized bid-weighted ordering. [53]

These mechanisms can be analyzed under one agent model.

Agents:

$$\{\text{users}, \text{searchers}, \text{builders}, \text{validators}, \text{LPs}\}.$$

Candidate mechanisms:

- priority-fee ordering,
- first-price bundle auction,
- MEV-sharing/backrun auction,
- batch auction,
- randomized ordering,
- PRECEDE-like power-weighted lottery,
- user-rebate mechanism,
- exclusive versus non-exclusive order flow.

Methodology

Calibrate an agent-based model using empirical distributions from historical blocks: trade sizes, arbitrage margins, bid/payment ratios, builder wins, swap price impact and volatility.

Run repeated games in which agents may switch routing providers and builders may pay for exclusive flow.

Test equilibria under different searcher populations, network latencies and risk preferences.

Metrics

User surplus; LP surplus; validator revenue; searcher profit; price convergence; transaction latency; block utilization; builder market share; HHI/Gini concentration; welfare; incentive-compatibility regret; fraction of exclusive order flow.

The main question should be Pareto-oriented:

> Can user welfare improve **without merely transferring an equivalent amount of cost into worse price discovery, lower validator participation, or higher builder concentration?**

Required skills

Game theory, mechanism design, simulation, Python/Rust, statistics and DeFi market microstructure.

Challenges

Results are highly assumption-sensitive. The strongest paper would therefore calibrate parameters from observed data and publish sensitivity ranges rather than claim one universal equilibrium.

Publication targets

ACM Economics and Computation, **ACM AFT**, **Financial Cryptography**, or security venues if the mechanism directly mitigates an attack class.

Overall assessment: high theoretical value; better as a second paper or PhD-style extension than as the first prototype.

Cross-Domain MEV Provenance

Research question. Can MEV be causally attributed to its originating market event rather than only the on-chain transaction that captures it?

Hypothesis.

> A temporal causal graph joining DEX state, L1/L2 transactions and external exchange prices will identify materially more arbitrage opportunity creation than atomic on-chain cycle detectors and will separate externally created price-discovery MEV from user-created toxic MEV.

Novelty

Recent work explicitly reframes MEV around opportunity creation rather than extraction. [29] The broader 2026 literature likewise emphasizes movement toward cross-chain MEV. [28]

Construct:

$$event \rightarrow state\ perturbation \rightarrow opportunity \rightarrow searcher\ observation \rightarrow execution \rightarrow hedge.$$

For each arbitrage, infer whether the opportunity was most plausibly caused by:

- a DEX user transaction,
- an oracle update,
- an L2/L1 state transition,
- a CEX price movement,

- another arbitrage,
- a liquidation,
- a bridge event.

Methodology

Use sub-second CEX market data, Ethereum/L2 blocks and pool states. Match opportunities using timestamp intervals and economic consistency constraints.

Apply event-study methods, structural causal models, temporal graph search or probabilistic attribution.

Metrics

Fraction of on-chain arbitrage with identifiable origin; attribution confidence; unexplained MEV; cross-domain latency; prediction of next arbitrage; estimated user-created versus exogenous MEV; price-discovery delay.

Required skills

Market microstructure, time-series econometrics, data engineering, blockchain tracing and causal inference.

Challenges

Clock synchronization, CEX data cost, non-public order flow, L2 sequencing timestamps and hidden hedging venues make definitive attribution difficult.

Publication targets

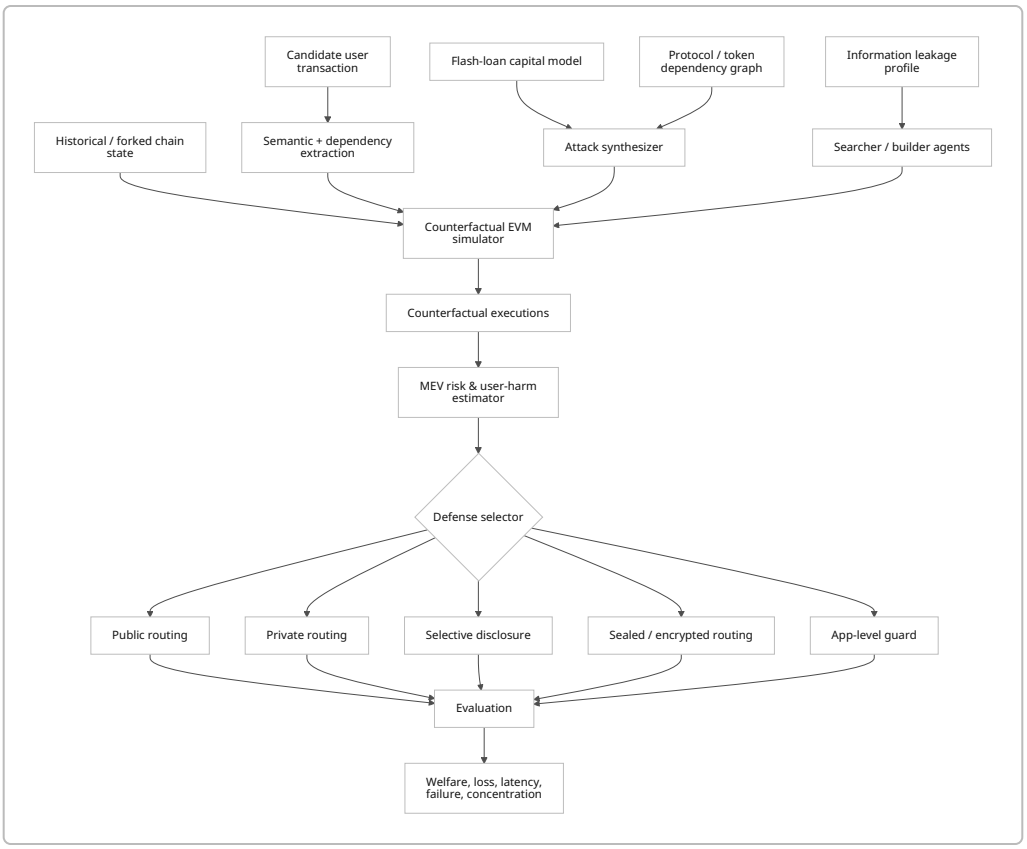
ACM IMC, USENIX Security, Financial Cryptography, ACM AFT, or quantitative-finance venues depending on emphasis.

Overall assessment: probably the most intellectually ambitious direction, but least suitable for a short first project.

Project Plan and Resources

Recommended integrated research architecture

The strongest overall project combines the first three directions:



The publishable contribution is the combination of **economic attack synthesis and defensive counterfactual evaluation**, not a profitable live arbitrage bot.

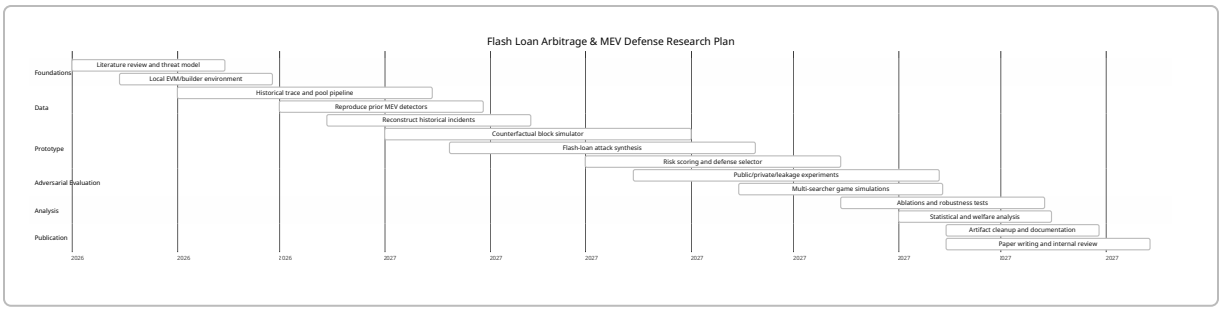
The system's key output for a transaction should resemble:

$$\begin{aligned} P(\text{sandwichable}) &= 0.83 \\ H_{95\%} &= \$41.70 \\ P(\text{private leakage attack}) &= 0.11 \\ C_{\text{protected route}} &= \$0.18 \\ \Delta L &= 140\ \text{ms} \\ \text{recommended defense} &= \text{protected route}. \end{aligned}$$

Those numbers are illustrative—not empirical findings—but show the type of measurable decision the system should eventually produce.

Suggested ten-month timeline

A ten-month schedule is long enough to produce both a systems artifact and a rigorous paper while leaving room for failed experiments.



The main **go/no-go milestone** should be around the end of month four. By then the project should reproduce at least one established detector and at least two historical economic attack scenarios on forked state. Failure to do that is evidence that the infrastructure scope is too large; the project should then narrow to counterfactual swap MEV rather than continuing to broaden protocol coverage.

Minimal research resources

Compute. No GPU is inherently necessary for the main project. Start with one high-memory workstation or cloud VM in the approximate class of 16–32 modern CPU cores, 64–128 GB RAM and fast NVMe storage. Reserve several terabytes if keeping extensive historical execution data locally. If storage or node maintenance becomes the bottleneck, an archive RPC provider plus locally cached traces is a reasonable research trade-off.

GPU resources become useful only if the project introduces graph neural networks, reinforcement-learning searchers or large learned transaction classifiers. They should not be a prerequisite for the first paper.

Chain environment. Use a fully isolated local devnet for adversarial experiments. Decker is currently Flashbots' recommended successor to the older Builder Playground, while `rbuilder` supports end-to-end block-builder testing and lists Sepolia and Hoodi connectivity in its current documentation. [54] Mainnet should primarily be used as a historical data source, not as an environment in which to run attack experiments.

Software stack.

- `rbuilder` for counterfactual block construction and backtesting. [31]
- Decker for reproducible isolated Ethereum networks. [38]
- Artemis if reusable searcher/strategy abstractions are desirable. [32]
- Foundry/Anvil for contract-level forking and test execution.
- Rust or Go for latency-sensitive systems code; Python for data science and optimization; Solidity for attack/defense contracts.

Core datasets. Ethereum canonical transaction/receipt/log/trace data should provide the base state. Flashbots' public Protect and MEV-Share datasets are especially valuable because they add visibility into historically private or selectively disclosed order flow that pure on-chain data cannot provide. [40] `rbuilder` also integrates with Flashbots' historical mempool data tooling for builder backtesting. [31]

For swap research, reconstruct pool states from on-chain events and compare against transaction-level execution. For cross-domain work, add high-resolution CEX order-book/trade data only after the on-chain system works; otherwise external-data engineering will dominate the project.

Experimental discipline

Every reported defense result should have at least four baselines:

$$\{\text{public}, \text{private}, \text{static rule}, \text{oracle/optimal hindsight}\}.$$

Every attack result should distinguish:

$$\text{observed extraction} \;/\; \text{maximum simulated extraction} \;/\; \text{realistically achievable extraction}.$$

That distinction prevents one of the most common conceptual errors in MEV research: treating a theoretically optimal reorder as though a searcher necessarily knew enough, quickly enough, to execute it.

A robust paper should also include **ablation studies**. Remove flash-loan access; hide transaction contents; vary searcher count; vary latency; remove CEX information; restrict builders; change fee levels; perturb liquidity; introduce stale state. A defense that works only with one exact historical configuration is an anecdote rather than a system result.

Finally, release the experimental harness and sanitized datasets wherever possible. Recent work on token-layer MEV provides a good ethical precedent: the researchers used historical simulations and responsible disclosure and deliberately withheld artifacts that would directly facilitate exploitation. [26] A flash-loan defense project can similarly publish detection and replay infrastructure while avoiding turnkey live exploit code.

Research Recommendation

Best primary thesis

The strongest single research question is:

> **Can we build a counterfactual, flash-loan-aware MEV risk engine that predicts the maximum plausible user loss of a pending transaction and automatically selects the cheapest defense that keeps that loss below a target bound?**

This question is strong because it unifies several disconnected literatures:

$$\text{attack synthesis} + \text{MEV measurement} + \text{block building} + \text{private-order-flow security} + \text{user welfare}.$$

It is also experimentally falsifiable.

For example:

$$H_0 : \text{Adaptive routing provides no material advantage over fixed private routing.}$$

$$H_1 : \text{Adaptive routing significantly reduces user loss at comparable cost and latency.}$$

The system need not "solve MEV" to produce a publishable contribution. Demonstrating that existing fixed defenses are systematically overprotective for some transactions and underprotective for others—and providing a calibrated decision mechanism—would itself be significant.

Priority by research payoff

Direction	Novelty	Feasibility	Data availability	Systems difficulty	Publication potential	Recommendation
Adaptive Counterfactual MEV Defense	Very high	**High**	High	Medium–high	**Very high**	**Start here**
Flash-Loan Attack Synthesis	High	Medium–high	High	High	**Very high**	Build as core module
Private/ Encrypted Leakage	**Very high**	Medium–high	Medium	High	**Very high**	Strong second study
MEV Market Mechanism Design	High	Medium	High	Medium	High	Theory extension
Cross-Domain Provenance	**Very high**	Medium–low	Medium–low	Very high	**Very high**	Longer-term/ high-risk

The best scoped first paper would focus on **swaps and arbitrage**, not every DeFi operation. Then demonstrate extensibility with one non-price flash-loan class such as governance or lending/oracle manipulation. This preserves enough generality for a meaningful security claim while keeping state reconstruction and ground truth manageable.

A plausible paper-level contribution set would be:

1. a formal definition of counterfactual MEV risk under information and capital constraints;
2. a flash-loan-aware adversarial block simulator;
3. an adaptive transaction-defense policy;
4. a new dataset of counterfactual executions derived from historical Ethereum state;
5. empirical comparison against public/private/static-routing baselines;
6. open-source reproducibility infrastructure.

That combination is substantially more novel than "we built a faster arbitrage bot" and directly targets the gap between current MEV extraction research and practical user protection.

Prioritized Sources

Essential foundational papers

Daian et al., "Flash Boys 2.0: Frontrunning, Transaction Reordering, and Consensus Instability in Decentralized Exchanges," IEEE S&P 2020. The conceptual starting point for the modern MEV literature: priority-gas auctions, arbitrage competition and consensus instability. arXiv. [5]

Qin et al., "Attacking the DeFi Ecosystem with Flash Loans for Fun and Profit," Financial Cryptography 2021. Essential for understanding flash loans as economic-attack amplifiers and for optimization-based incident reconstruction. arXiv. [3]

Zhou et al., "On the Just-In-Time Discovery of Profit-Generating Transactions in DeFi Protocols," IEEE S&P 2021. DEFIPOSER is the closest classic predecessor to an automated flash-loan/MEV opportunity synthesizer. arXiv. [20]

Qin, Zhou & Gervais, "Quantifying Blockchain Extractable Value: How Dark Is the Forest?", IEEE S&P 2022. Large-scale measurement of arbitrage, sandwiching, liquidations and generalized front-running. arXiv. [14]

Babel et al., "Clockwork Finance: Automated Analysis of Economic Security in Smart Contracts," IEEE S&P 2023. The strongest foundational source for compositional, formally defined economic security. arXiv. [23]

McLaughlin, Kruegel & Vigna, "A Large Scale Study of the Ethereum Arbitrage Ecosystem," USENIX Security 2023. Strongest broad empirical baseline for Ethereum arbitrage: 28 months, 3.8 million detected arbitrages and billions of candidate opportunities. USENIX. [24]

Essential MEV-defense and market-structure papers

Zhou et al., "High-Frequency Trading on Decentralized On-Chain Exchanges." Analytical and empirical foundation for sandwich attacks. arXiv. [13]

Bartoletti et al., "Maximizing Extractable Value from Automated Market Makers." Formal baseline for constructing optimal adversarial AMM strategies. arXiv. [21]

Weintraub et al., "A Flash(bot) in the Pan: Measuring MEV in Private Pools." Essential evidence that public-mempool-only measurement misses a large share of the MEV market. arXiv. [8]

Wahrstätter et al., "Time to Bribe: Measuring Block Construction Markets." Key post-Merge/PBS empirical work. arXiv. [25]

"Don't Let MEV Slip: The Costs of Swapping on the Uniswap Website." Valuable because it evaluates MEV from the **user execution-quality** perspective rather than only searcher revenue. arXiv. [19]

Current research frontier

“SoK: The Evolution of Maximal Extractable Value, From Miners to Cross-Chain,” 2026. Useful map of the transition from miner-era MEV toward cross-domain and cross-chain problems. arXiv. [28]

“More to Extract: Discovering MEV by Token Contract Analysis,” 2026. Particularly relevant to a new security-analysis system because it demonstrates that token semantics expose MEV missed by application-layer tools. arXiv HTML. [26]

Chen et al., “Time Is Money: Incentivized Causal Transaction Ordering,” 2026. A new game-theoretic approach to causality/front-running through randomized bid-weighted ordering. arXiv HTML. [27]

“The Origins of MEV: Systematic Attribution of Arbitrage Opportunity Creation,” 2026. Important for the proposed cross-domain provenance research because it distinguishes creation of an opportunity from eventual extraction. [29]

Protocol and industry primary sources

Ethereum, “Maximal extractable value.” Canonical definition and overview of Ethereum's MEV problem. Ethereum.org. [2]

Ethereum, “Proposer-builder separation.” Current Ethereum roadmap explanation of PBS and its relationship to MEV. Ethereum.org. [55]

EIP-7732, “Enshrined Proposer-Builder Separation.” Primary specification for the proposed in-protocol PBS mechanism. EIP-7732. [17]

EIP-7805, “Fork-choice Enforced Inclusion Lists.” Primary current proposal for committee-enforced censorship resistance against concentrated builders. EIP-7805. [41]

EIP-8184, “LUCID encrypted mempool.” Especially valuable for identifying the unresolved engineering and cryptographic requirements of protected public order flow. EIP-8184. [42]

Aave V3 Flash Loan documentation. Canonical implementation documentation for the flash-loan primitive. Aave. [1]

Flashbots MEV-Boost / rbuilder / BuilderNet. Primary references for the production builder ecosystem and its ongoing evolution toward TEE-backed collaborative building. MEV-Boost repository, rbuilder repository, Flashbots. [56]

Security incidents and policy

Beanstalk, “Beanstalk Governance Exploit.” Rare first-party account demonstrating a flash-loan-enabled governance exploit rather than a conventional AMM-only attack. Beanstalk. [4]

Flashbots, April 2023 relay exploit postmortem. Essential reminder that relays/builders themselves become high-value security components in an MEV-defense architecture. [44]

UK Financial Conduct Authority, "Review of Maximal Extractable Value and Blockchain Oracles." One of the most useful regulator-produced starting points for an MEV policy research component; the FCA describes MEV as a source of technical risk and uses the work to develop regulatory understanding. FCA research note. [50]

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[3] [15] [52] Attacking the DeFi Ecosystem with Flash Loans for Fun and Profit
https://arxiv.org/abs/2003.03810?utm_source=chatgpt.com

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[8] A Flash(bot) in the Pan: Measuring Maximal Extractable Value in Private Pools
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